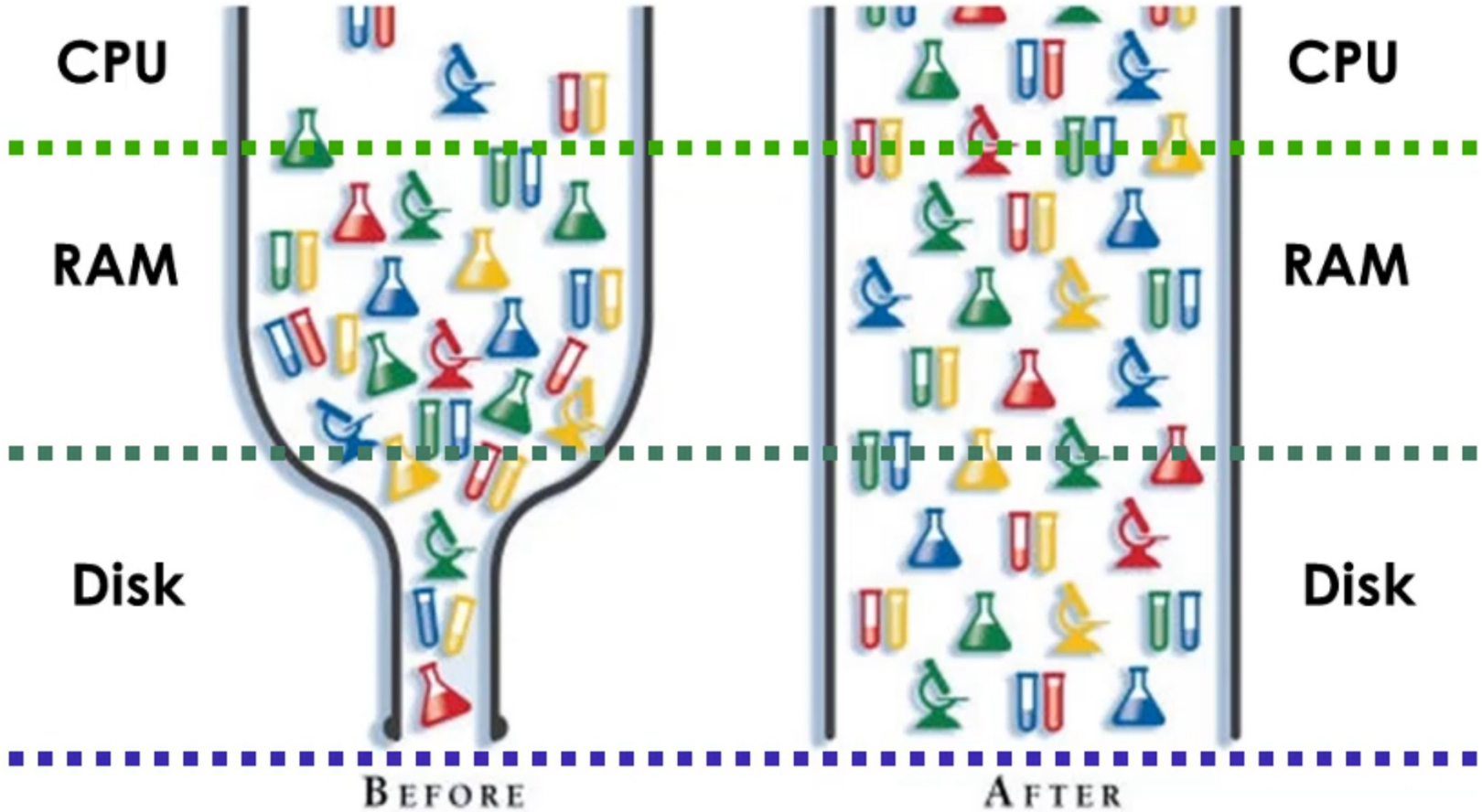
Practical LPIC-2

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Wait Time + Average Load



stress vs stress-ng

stress

- Simple workload generator
- CPU stress
- Memory stress
- I/O stress
- Disk stress

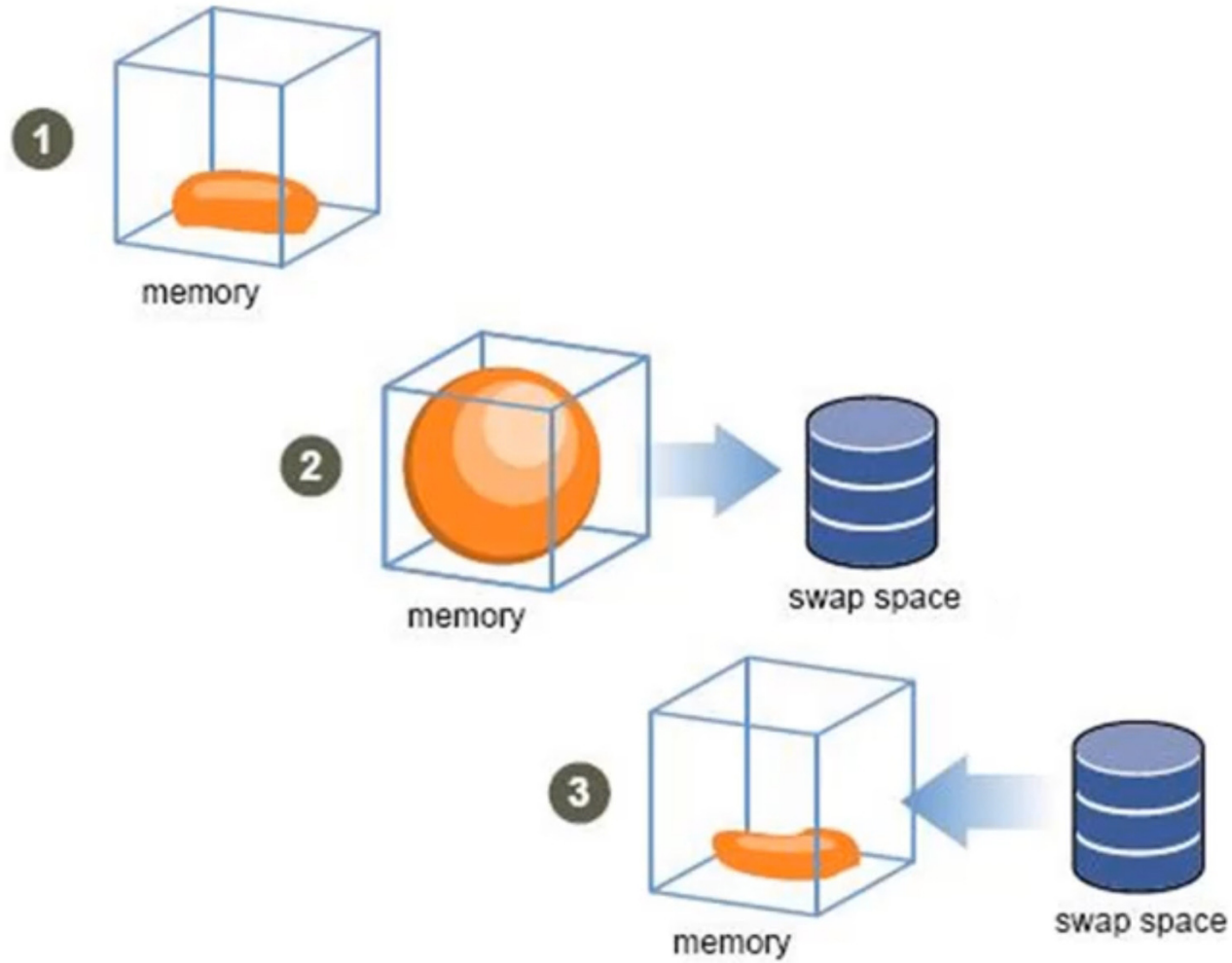
stress-ng

- Updated version of stress
- CPU compute
- Cache thrashing
- Drive stress
- I/O syncs
- VM stress
- Socket stressing
- Context switching
- Process creation and termination
- 60 different stress test
- 20 virtual memory stress test
- …

Should know commands :

- vmstat
- free
- top
- stress-ng (apt-get install stress-ng)

What is SWAP space ?



Disk IOPS commands

Should know commands :

- uptime
- sar
- iotop
- iostat
- lsof
- stress-ng –hdd
- apt-get install sysstat

What are the solution :

- Disk type (RPM)
- RAID
- SSD

Average System Load

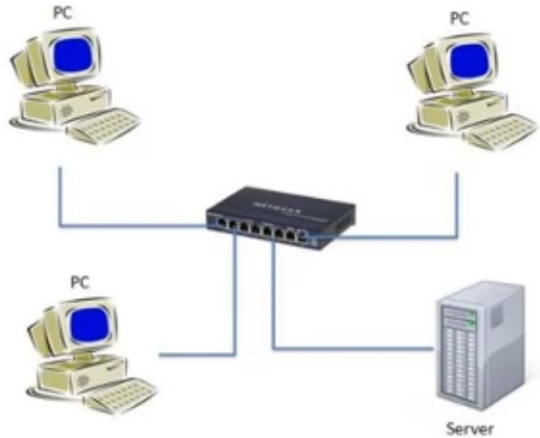
1CPU	4CPU	8CPU
0.5 = > Half Used	0.5 = > 12.5 %	?
1.0 = > Fully Used	1.0 = > 25 %	?
1.5 = > Overused	5.0 = > 125 %	?

load average: 0.00, 0.00, 0.00

Network I/O measures

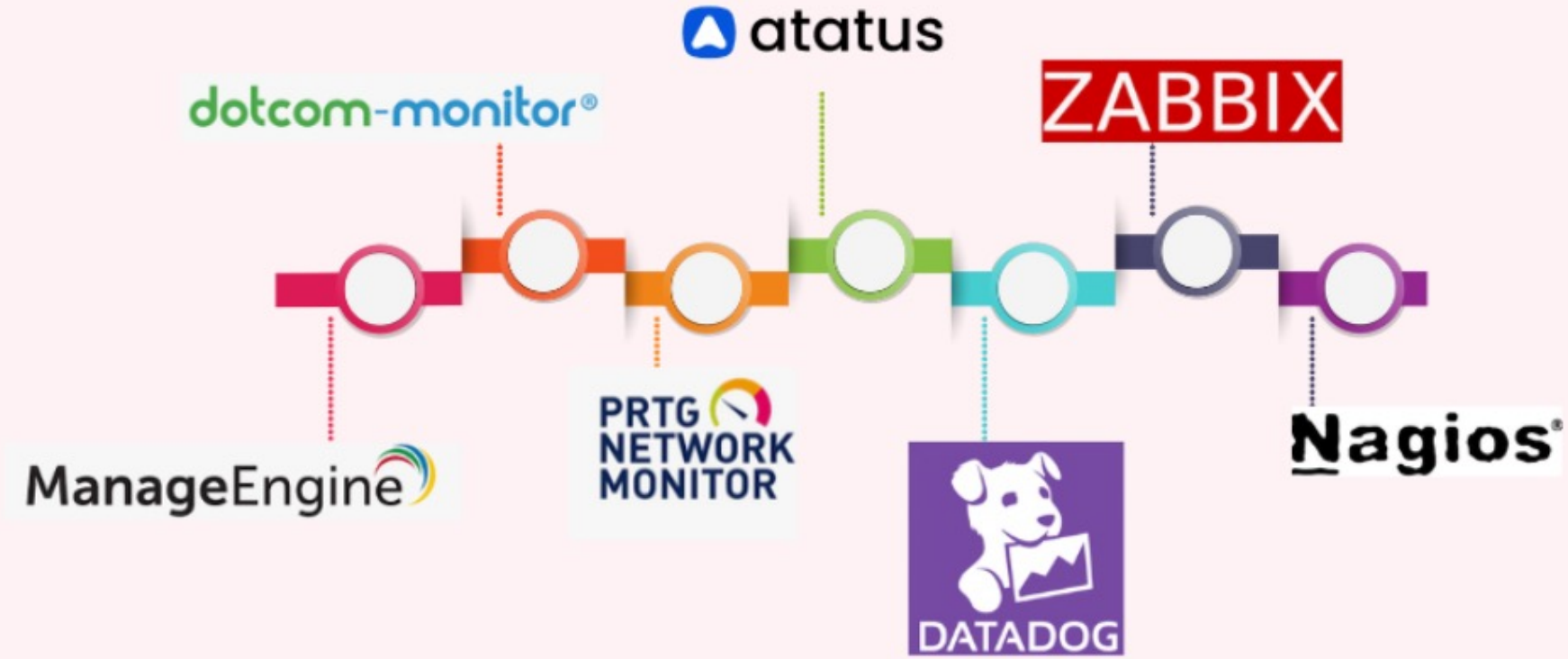
Should know :

- **How to identify interfaces ?**
- **How to see historical stats ?**
- **How to check realtime stats ?**
- **netstat command** (-ie) (-s) (-tuna)
- **Interface Top Users = iftop**
- **Network Load = nload =** bandwidth graph



مشکل از شبکه داخلی هست یا از شبکه خارجی ؟




atatus
dotcom-monitor
ZABBIX
ManageEngine
PRTG NETWORK MONITOR
DATADOG
Nagios

Kernel Version Numbering



Kernel Version Numbering

قبل از نسخه 2.6.x





عدد فرد یا odd نسخه Developer

عدد زوج یا even نسخه Stable

بعد از نسخه 2.6.x







حرف X یعنی همه نسخه ها و اصلاحیه ها

همگی از این به بعد Stable هستند

نسخه Developer یا Development

دیگر وجود ندارد

Longterm

Manipulating Kernel Modules

/lib/modules = > location of installed and available modules
uname –r = > viewing version of current kernel
lsnmod = > installing kernel modules with absolute path
rmmod = > removing kernel modules
modprobe = > managing kernel modules with dependencies
depmod = > creates and updates modules.dep
modinfo == > information about a module

Kernel automation and configuration

1. **sysctl.conf** = > Permanent configuration file for loading kernel modules
2. **/proc/sys/kernel** = > current live kernel modules
3. **/etc/sysctl.d/** = > packages loaded on kernel modules (sysctl.conf overrides this)
4. **/sbin/sysctl** (command)= > manipulating current kernel modules
5. **udev=** > Monitoring live adding and removing kernel modules
6. **lsusb**= > lists connected USB devices
7. **lspci** = > lists connected PCI devices
8. **lsdev** = > lists all connected system devices
9. **dmesg** = > running log of system devices + what has been done
10. **udevadm monitor** =>real-time monitoring of Kernel modules (old = udevmonitor)
11. **blacklist.conf** = > does not let udev to install old kernel modules

Linux Compilation Process



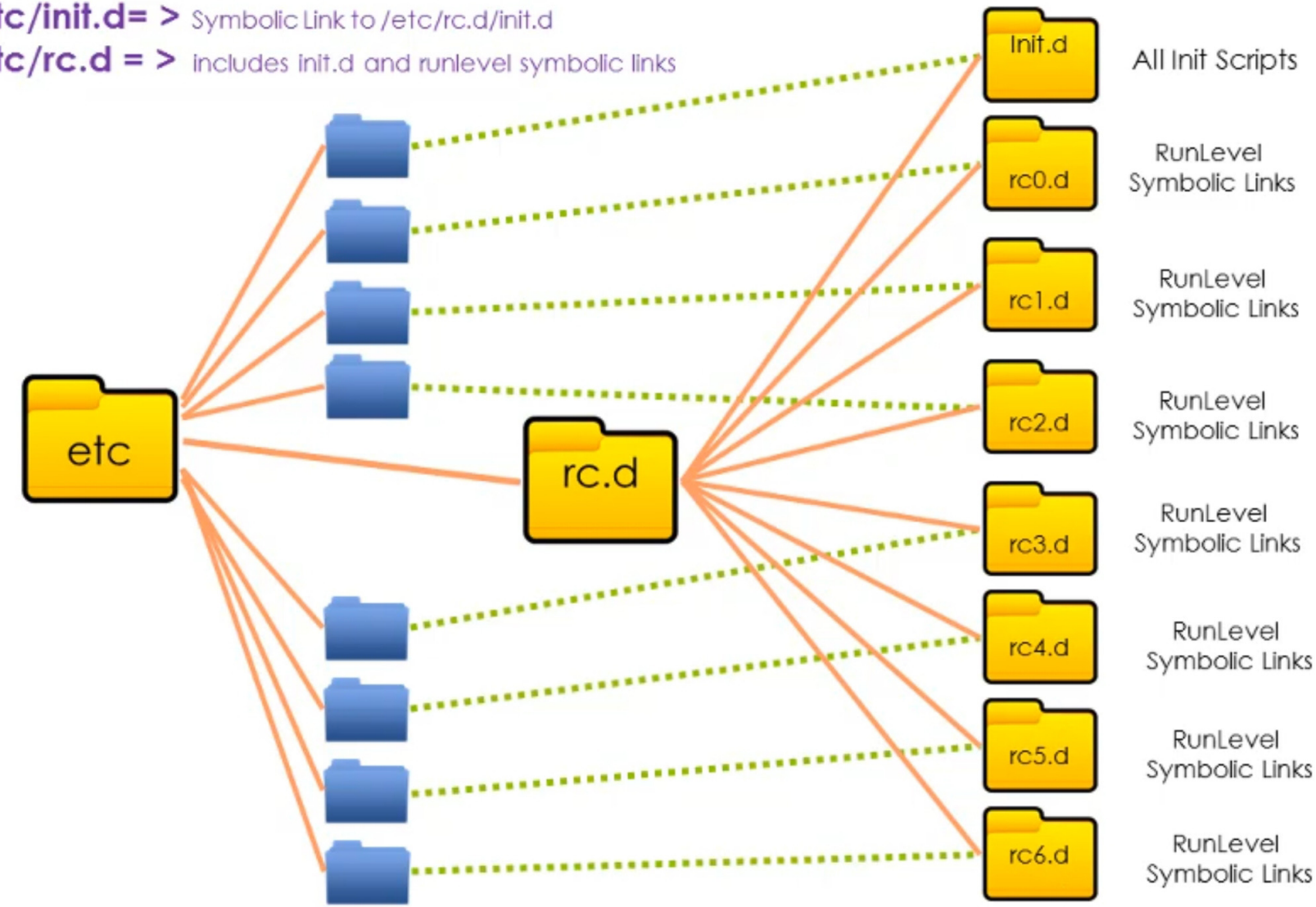


zImage = 640 KB
bzImage = More

SystemV Init Scripts

/etc/init.d= > Symbolic Link to /etc/rc.d/init.d

/etc/rc.d = > includes init.d and runlevel symbolic links





- SysV is serial processing
- SysV is slower
- Upstart is backward compatible
- Better support for hotplug devices
- Clearer and faster service management
- Upstart and Systemd are not in LPIC exam yet

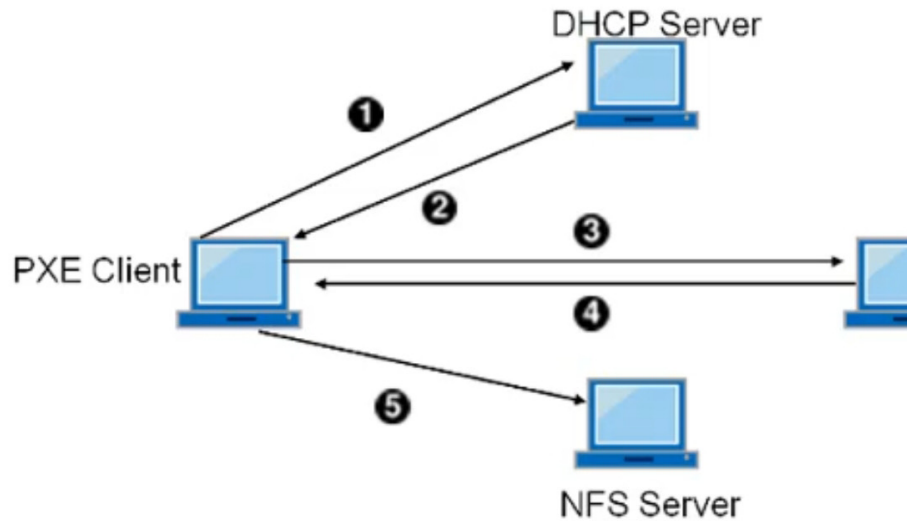
systemctl	start	servicename
systemctl	stop	servicename
systemctl	enable	servicename
systemctl	disable	servicename
systemctl	restart	servicename
systemctl	reload	servicename
systemctl	list-unit-files	--type=service

Alternate Bootloaders

- **LILO** = Linux Loader
- **syslinux** = USB Drives
- **extlinux** = Hard Drives
- **isolinux** = CD-ROM – DVD-ROM – ISO Image



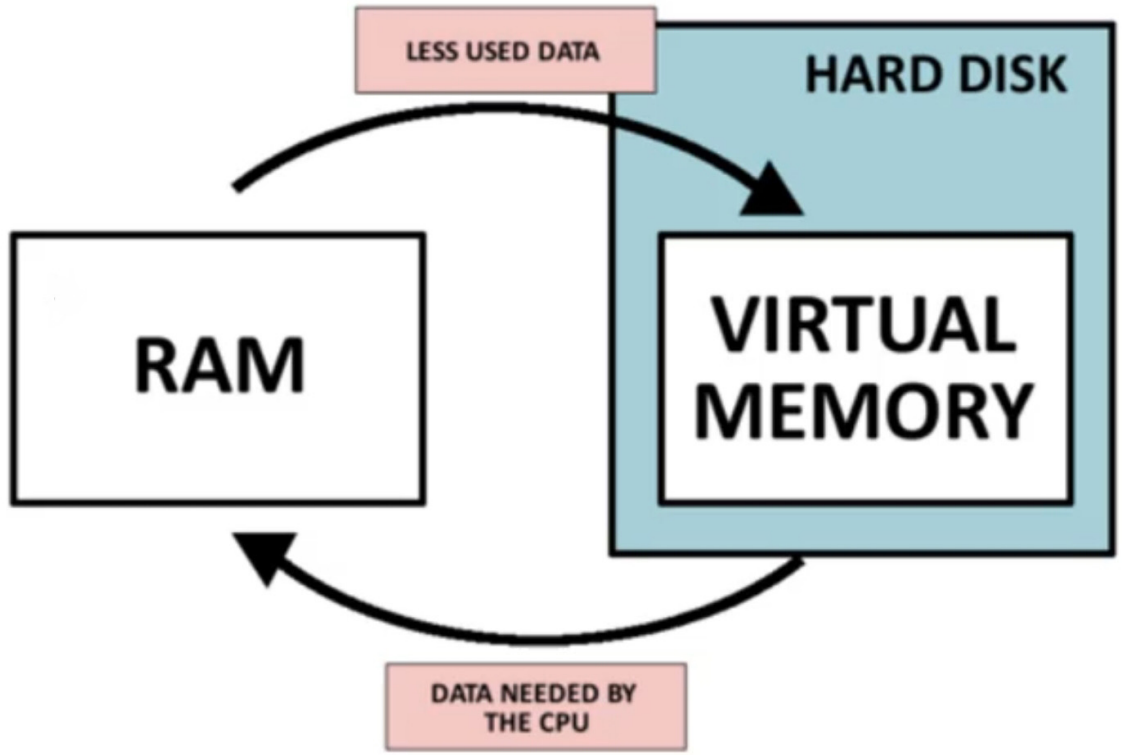
PXE = Pre-execution Environment



TFTP (NFS) Content : /var/lib/tftpboot

---pxelinux.0 (Download Kernel Loader)

---pxelinux.cfg → default (items)


LESS USED DATA
HARD DISK
RAM
VIRTUAL MEMORY
DATA NEEDED BY THE CPU

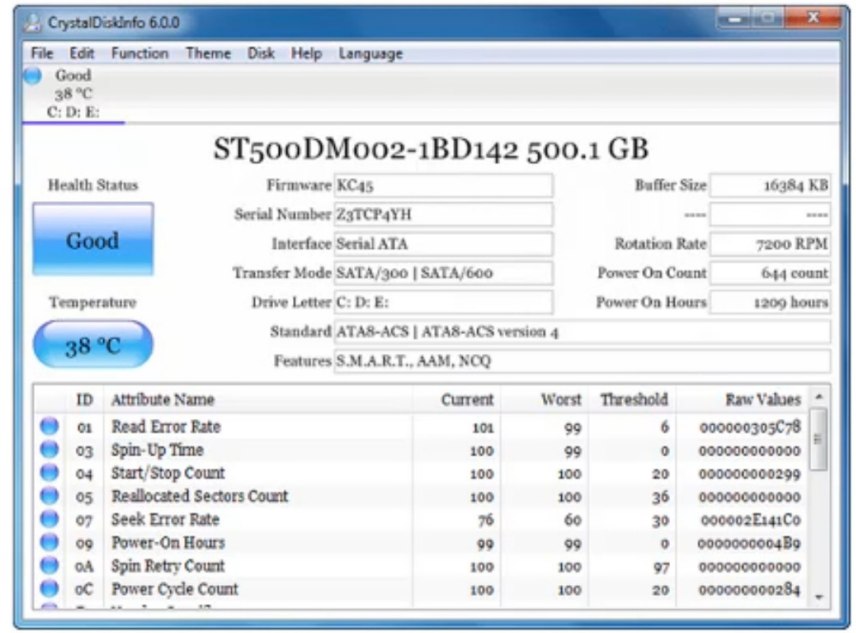
Self = S

Monitoring = M

Analysis = A

Reporting = R

Technology = T



smartctl

smartd

- **ISO9660 (Joliet , El-Torino , Rock Ridge)**
- **UDF (Universal Disk Format)**

- **Joliet** = long file names
- **El-Torino** = bootable CD
- **Rock Ridge** = Write Permissions on CD
- **UDF** = CD + DVD

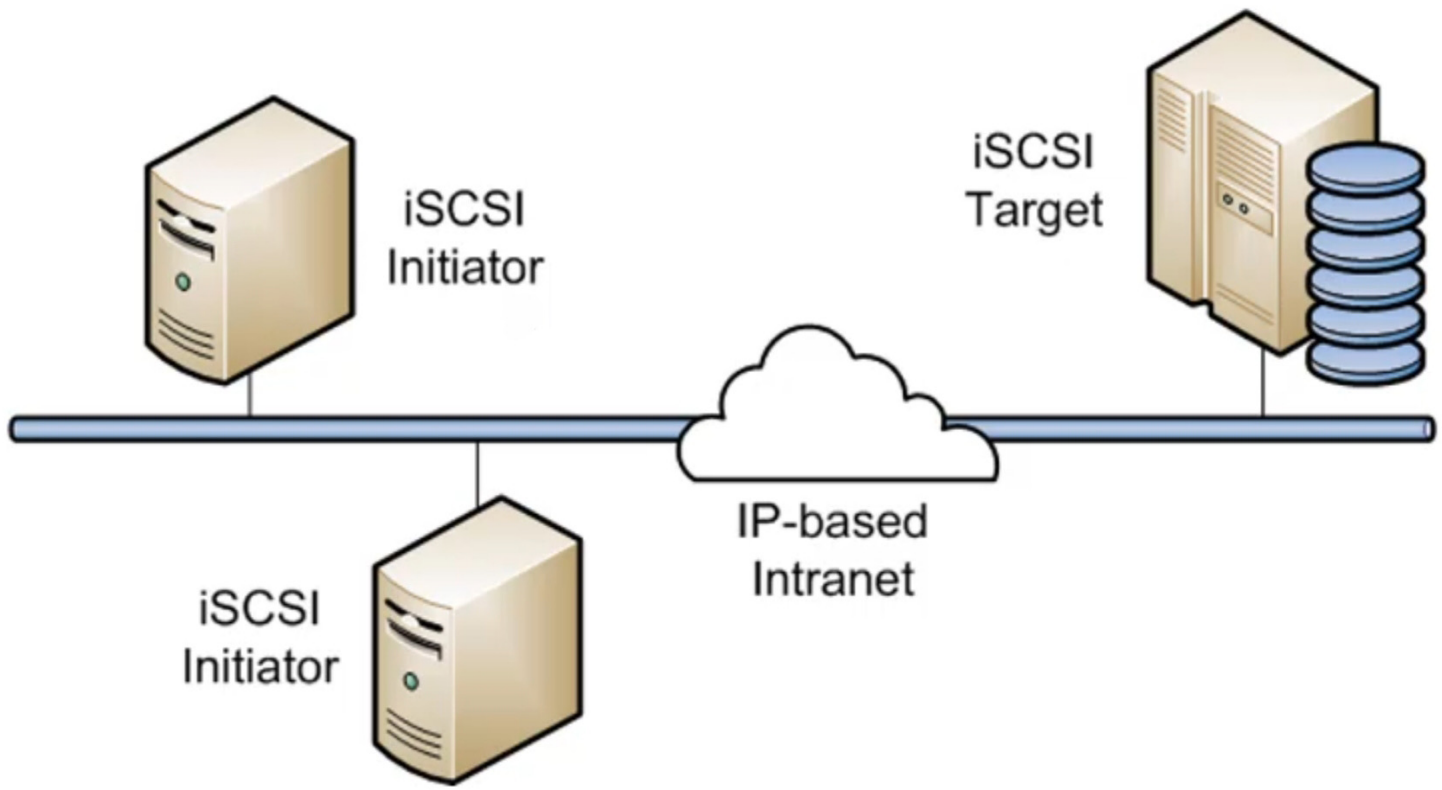


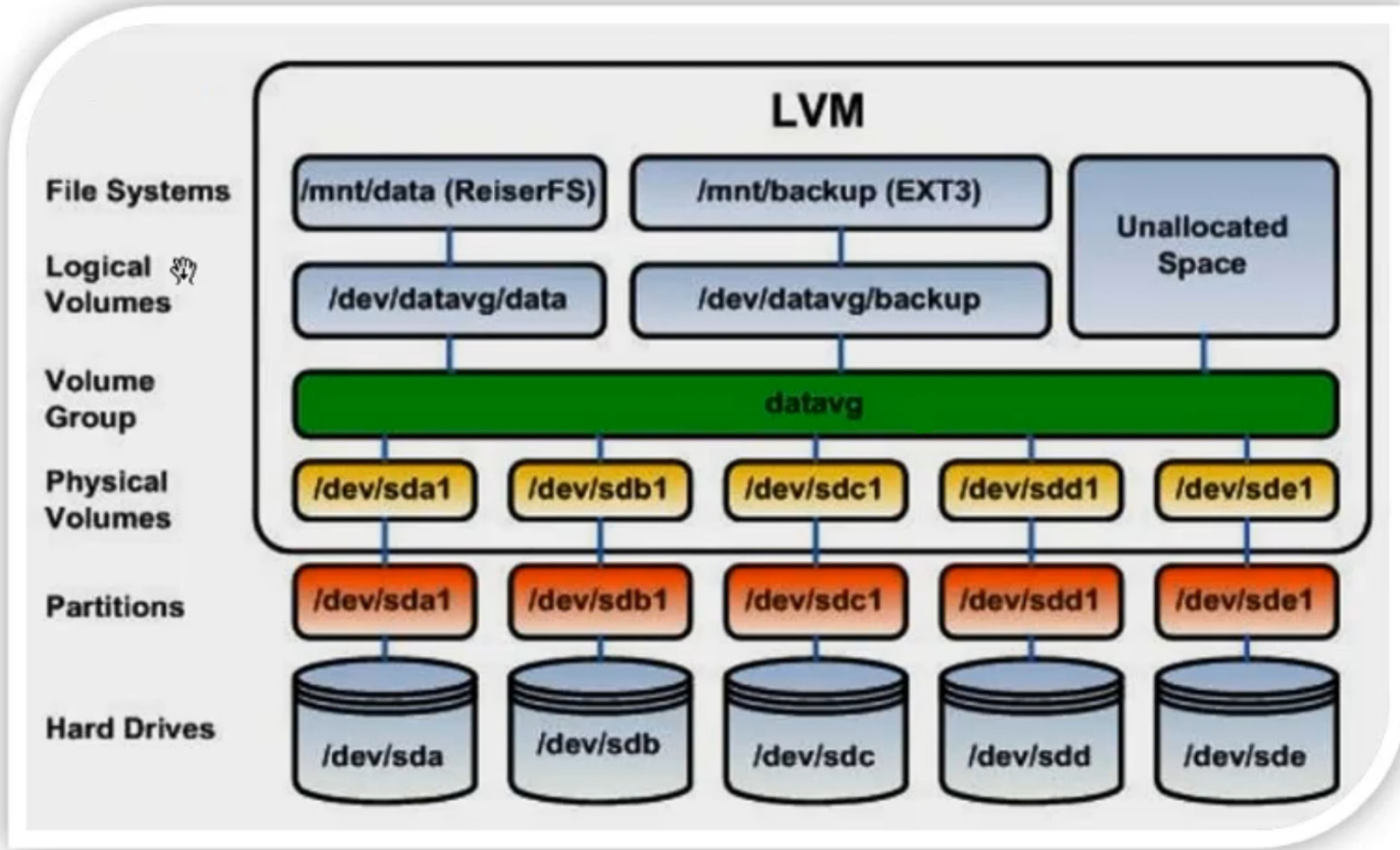
hdparm = /dev/had = IDE
sdparm = /dev/sda = SATA = SAS

- These are hard disk level tools

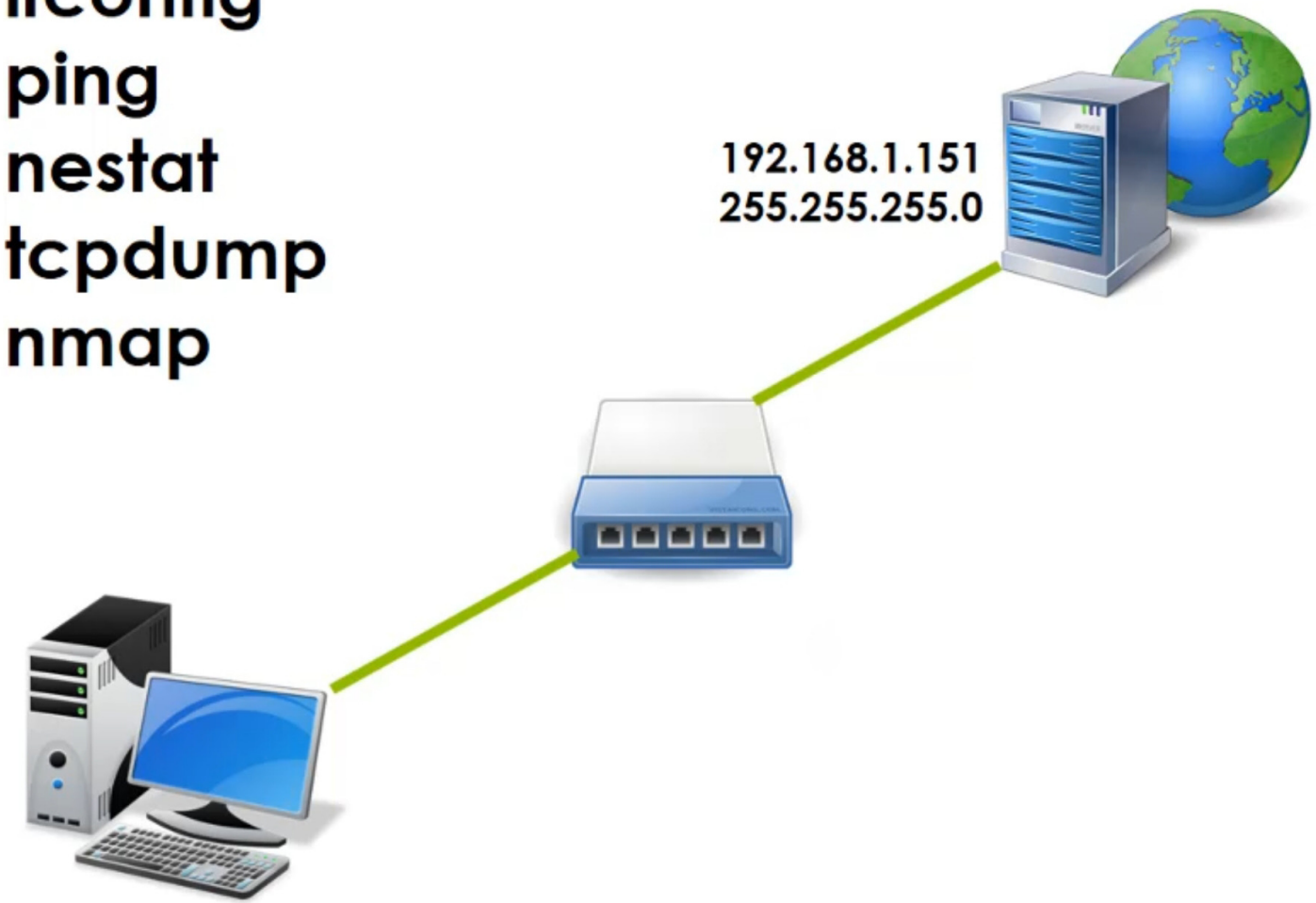
tune2fs = Tools at file system level
sysctl = Change settings in proc file system (interrupts and cpu ..)

- **iSCSI =** Protocol
- **LUN =** Logical Unit Number
- Disk Based Storage
- **iSCSI** Target
- **iSCSI** Initiator




LVM
File Systems
Logical Volumes
Volume Group
Physical Volumes
Partitions
Hard Drives
/mnt/data (ReiserFS)
/mnt/backup (EXT3)
Unallocated Space
/dev/datavg/data
/dev/datavg/backup
datavg
/dev/sda1
/dev/sdb1
/dev/sdc1
/dev/sdd1
/dev/sde1
/dev/sda1
/dev/sdb1
/dev/sdc1
/dev/sdd1
/dev/sde1
/dev/sda
/dev/sdb
/dev/sdc
/dev/sdd
/dev/sde

- ifconfig
- ping
- nestat
- tcpdump
- nmap



1) Source Code
2) Compile .deb
3) Advantages
4) Disadvantages
5) dependency Problems
6) ./configure
7) make
8) make install
9) searching from cache



What to backup ?

- /etc/
- /var/spool/
- /home/
- /usr/local/

Backup Packages

- Amanda
- Bacula
- BackupPC

What to not backup ?

- /usr/

What to may backup ?

- /var/lib/
- /var/log/
- /opt/

Tapes

- /dev/st0
- /dev/nst0
- /usr/bin/mt

Practical LPIC-1

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